Predicting aboveground biomass density according to LiDAR product within BIOMASS R package

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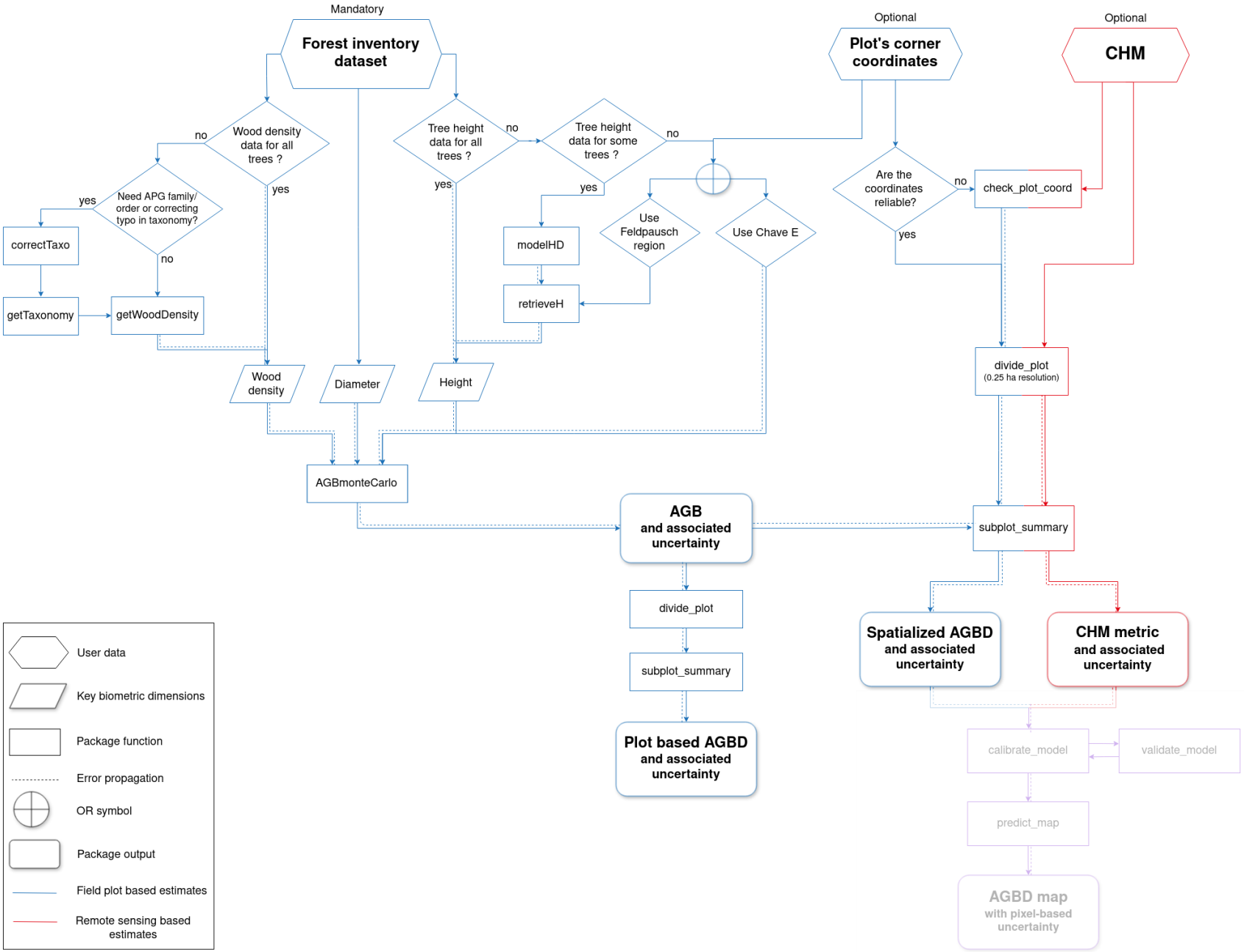
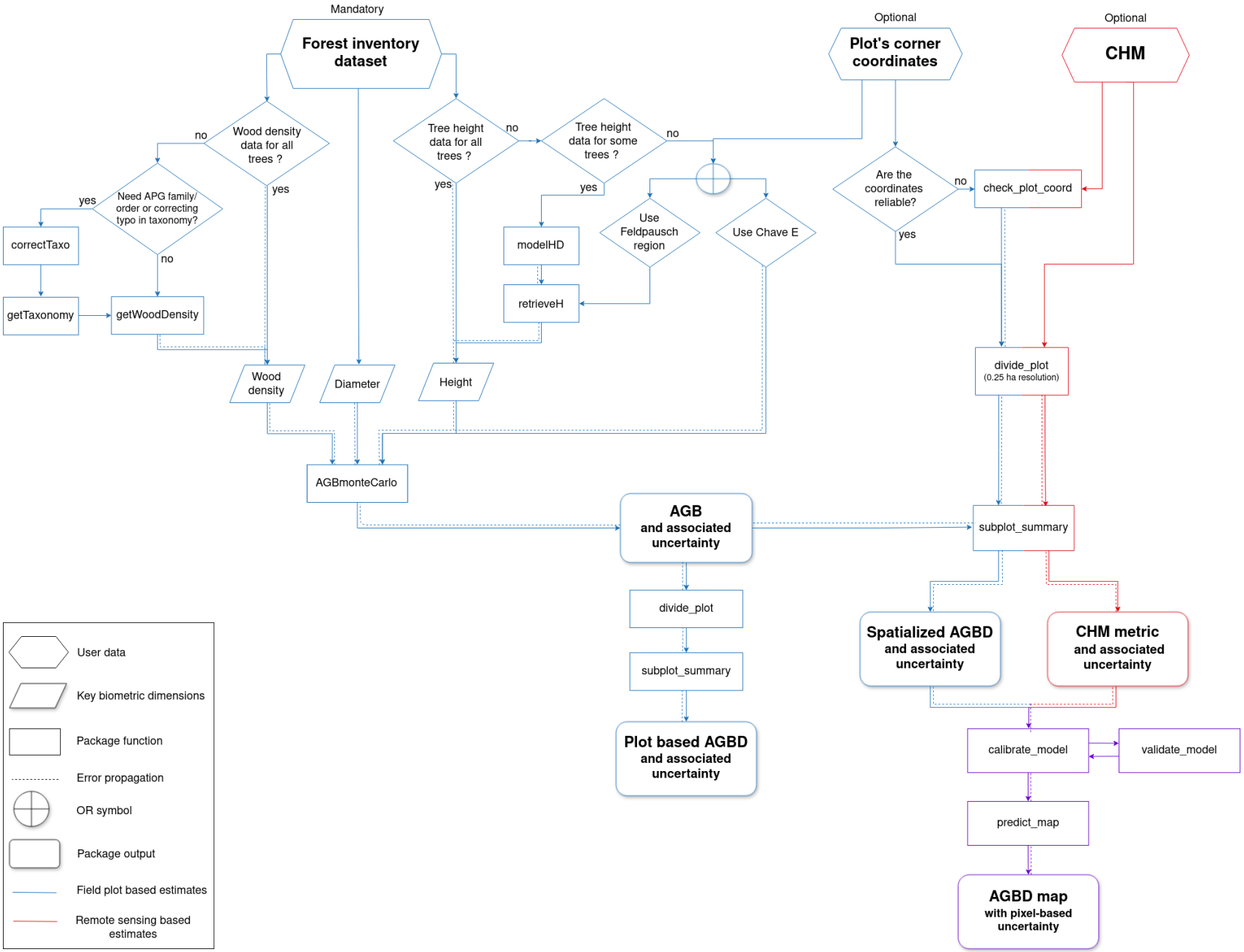
2026-01-30

BIOMASS R package FRM4BIOMASS



CHM-AGBD model calibration (V3)



CHM-AGBD model calibration: proposed statistical framework

- geostatistical model with SPV-I/C (SPatially Varying Intercept/Coefficients) to integrate spatial correlation:
- $y(s) = (\alpha + \tilde{\alpha}(s)) + (\beta + \tilde{\beta}(s)) \times x(s) + \epsilon(s)$
  with $\tilde{\alpha}(s_1), ..., \tilde{\alpha}(s_n) \sim MVN(0, C_\alpha(s_i, s_j))$
- references

Remote Sensing of Environment 169 (2015) 113–127

Contents lists available at ScienceDirect

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journal homepage: www.elsevier.com/locate/rse

LiDAR based prediction of forest biomass using hierarchical models with spatially varying coefficients

Chad Babcock [a], Andrew O. Finley [b,*], John B. Bradford [c], Randall Kolka [d], Richard Birdsey [e], Michael G. Ryan [f]

Remote Sensing of Environment 318 (2025) 114557

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

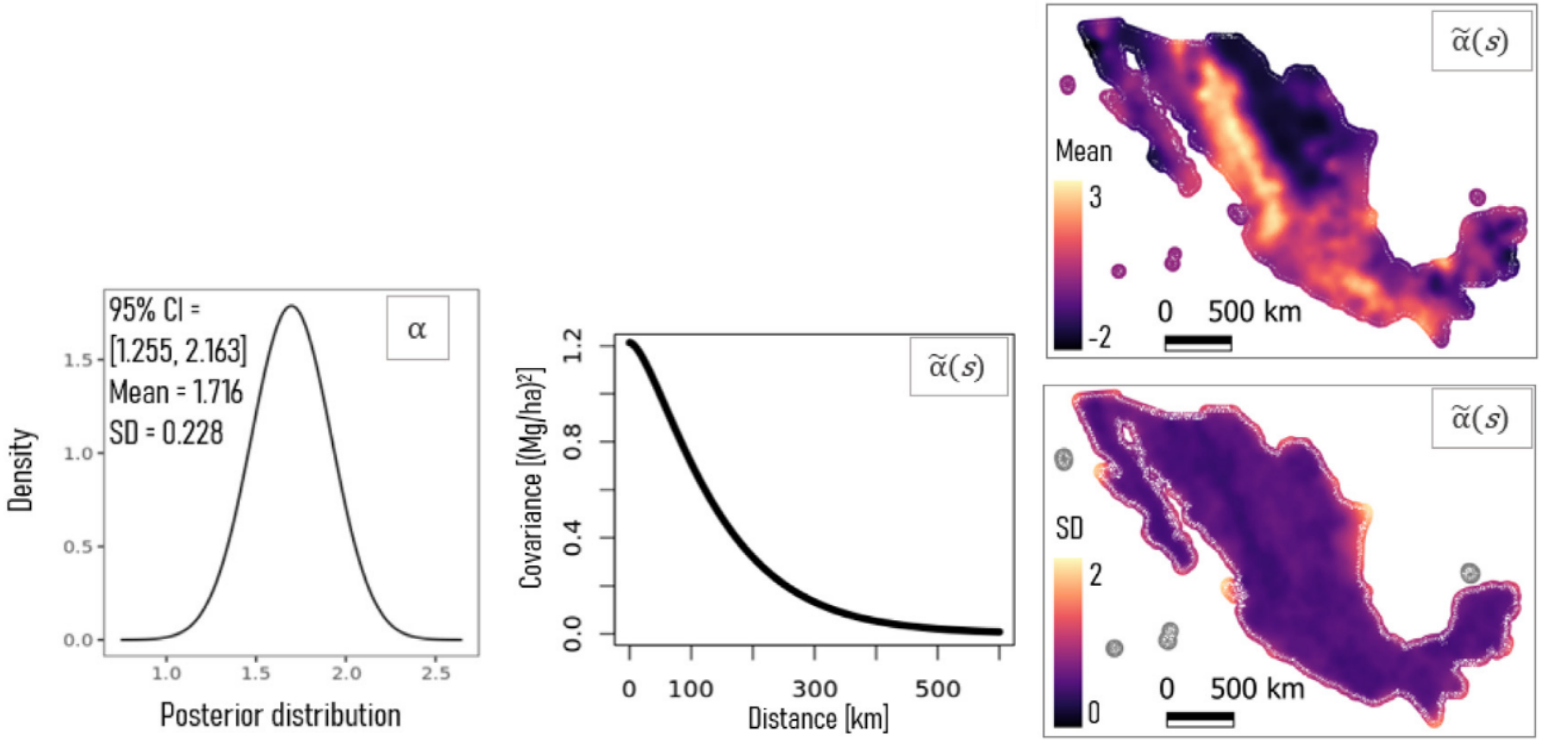
A geostatistical approach to enhancing national forest biomass assessments with Earth Observation to aid climate policy needs

Neha Hunka [a,*], Paul May [b], Chad Babcock [c], José Armando Alanís de la Rosa [d], Maria de los Ángeles Soriano-Luna [d], Rafael Mayorga Saucedo [d], John Armston [a], Maurizio Santoro [e], Daniela Requena Suarez [f], Martin Herold [f], Natalia Málaga [f], Sean P. Healey [g], Robert E. Kennedy [h], Andrew T. Hudak [i], Laura Duncanson [a]

CHM-AGBD model calibration: proposed statistical framework

$$y(s) = (\alpha + \tilde{\alpha}(s)) + (\beta + \tilde{\beta}(s)) \times x(s) + \epsilon(s)$$

with $\tilde{\alpha}(s_1), ..., \tilde{\alpha}(s_n) \sim MVN(0, C_\alpha(s_i, s_j))$



CHM-AGBD model calibration: proposed statistical framework

- SPDE procedure
- semi-automated workflow in a github repository
- INLA based no R package on CRAN

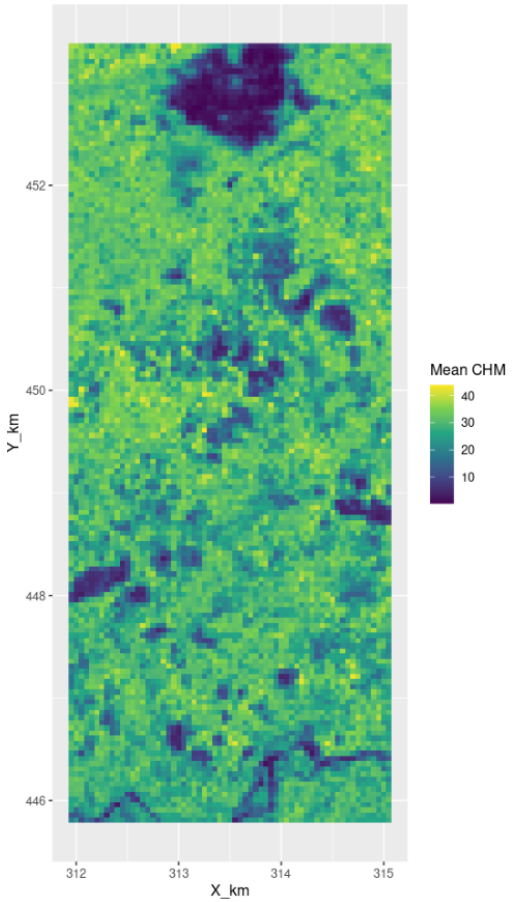
$\rightarrow$ need for our own workflow

Implementation challenges

- taking into account AGBD uncertainties
- computation time
- packages on CRAN

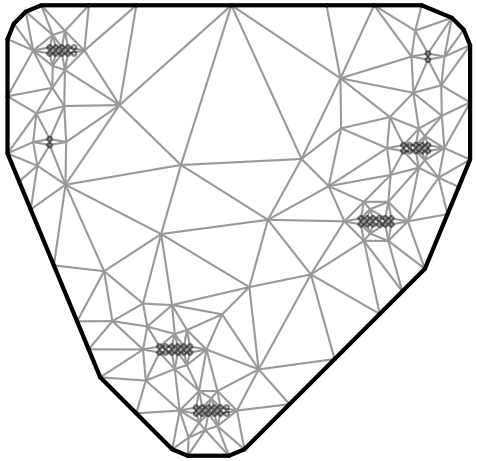
→ we chose brms R package

CHM-AGBD model calibration: example with simulated data

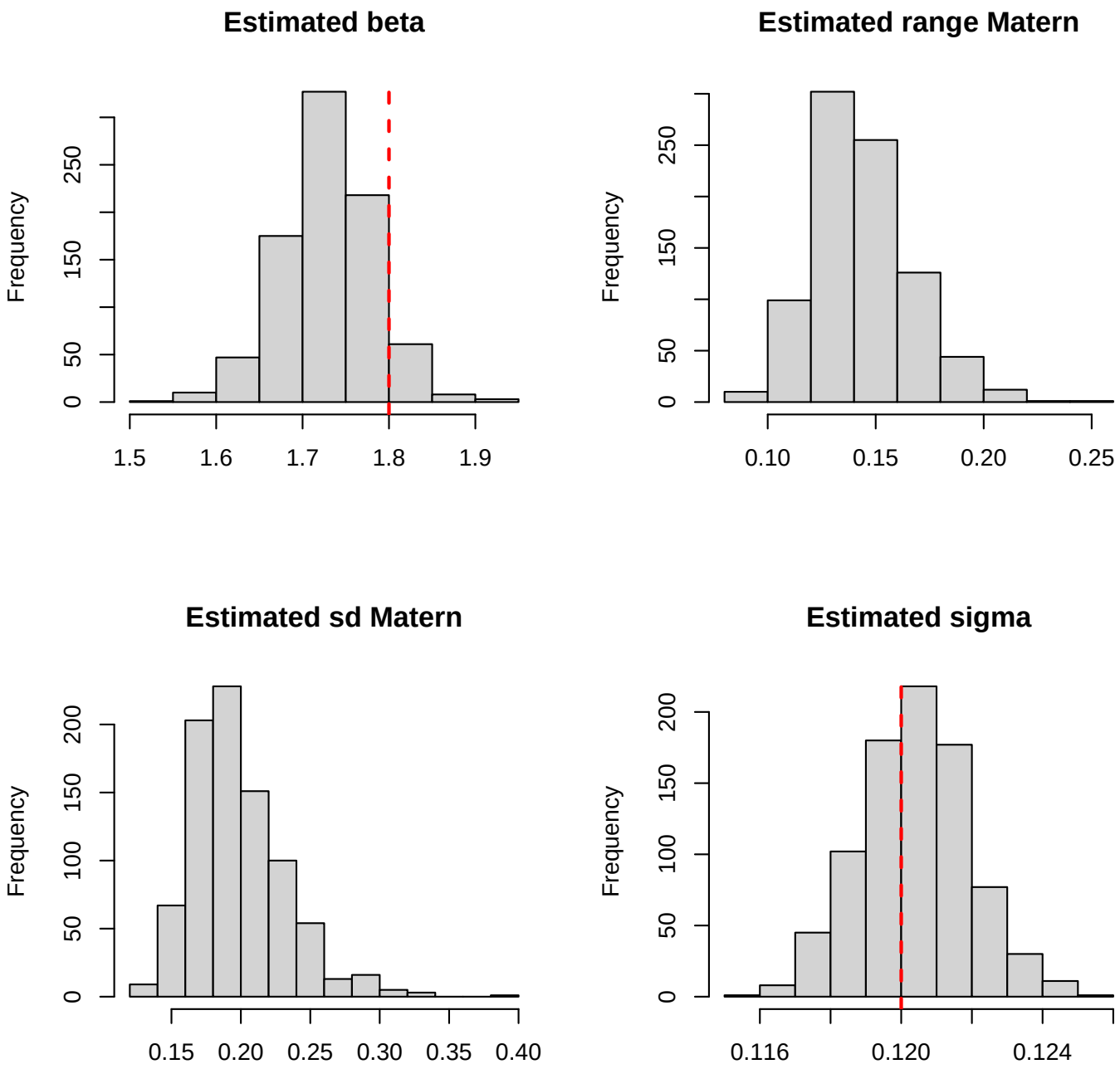


- SpVC model
  $log(AGBD(s)) =$
  $\alpha + (\beta + \tilde{\beta}(s)) \times log(CHM(s)) + \epsilon(s)$
  with
  $\tilde{\beta}(s_1), ..., \tilde{\beta}(s_n) \sim MVN(0, C_\beta(s_i, s_j))$
  and $\alpha = 0$

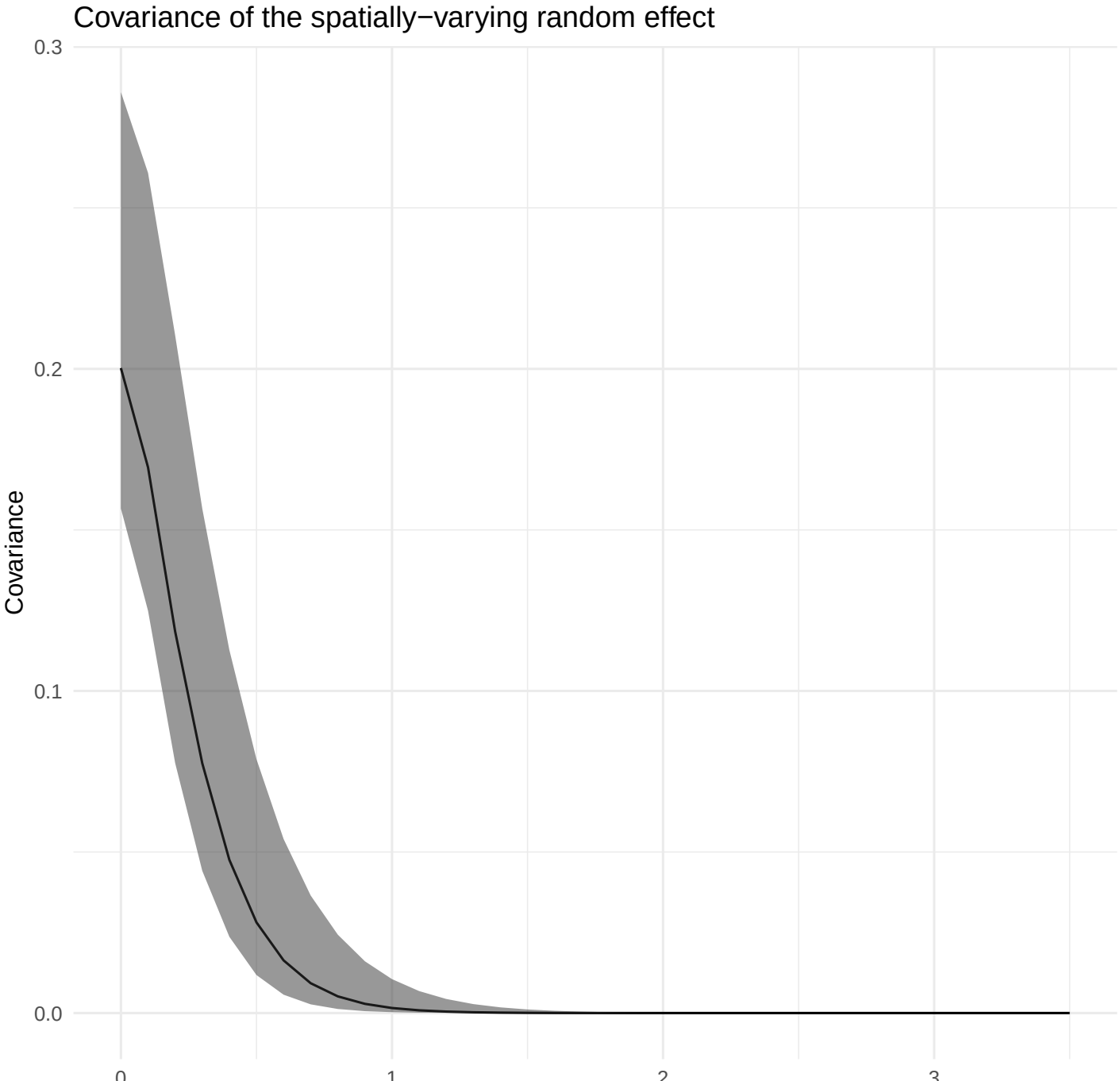
- Simulated data



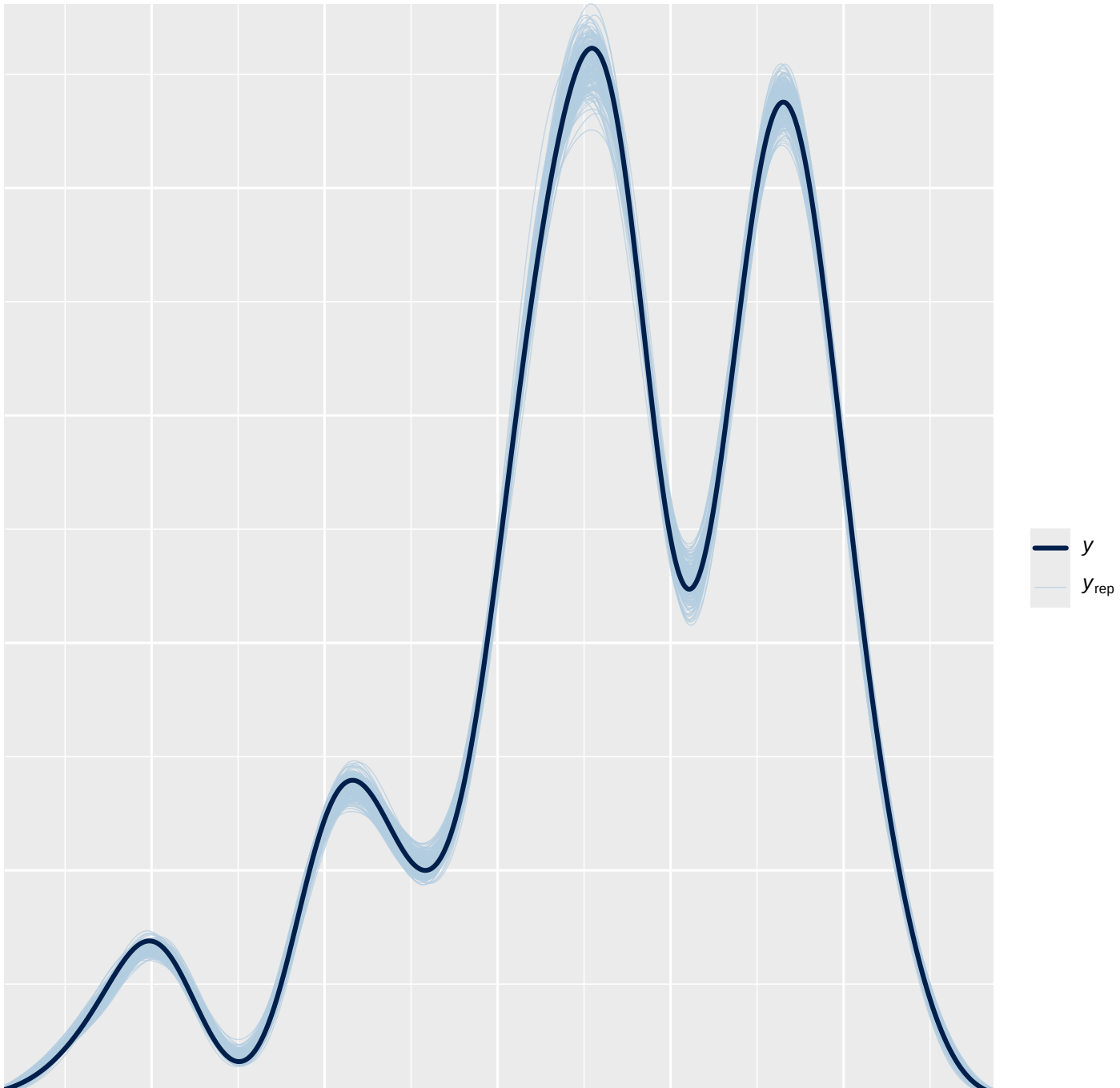
CHM-AGBD model estimates brms



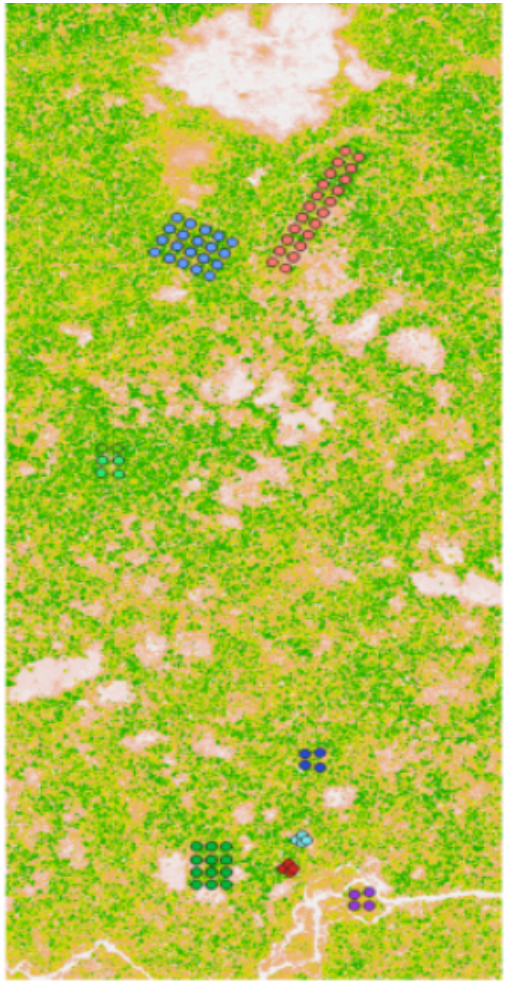
CHM-AGBD covariance functions



CHM-AGBD posterior predictive checks

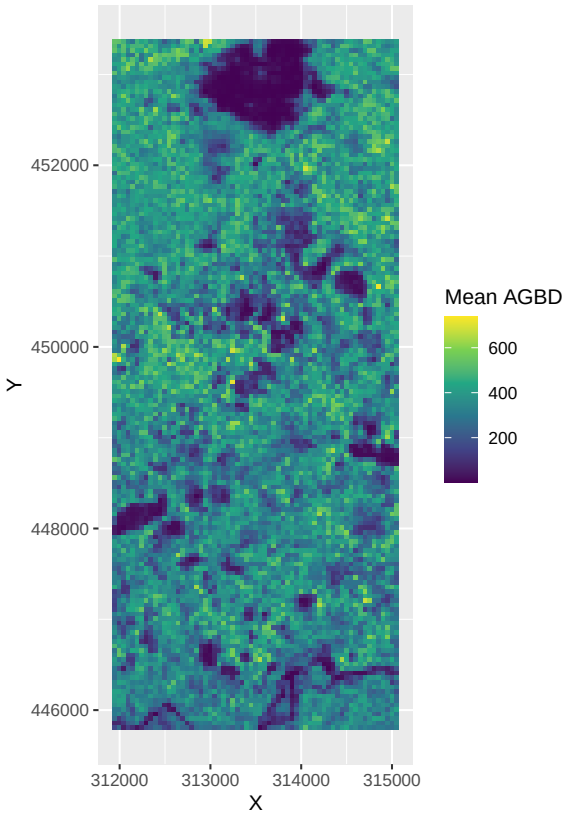


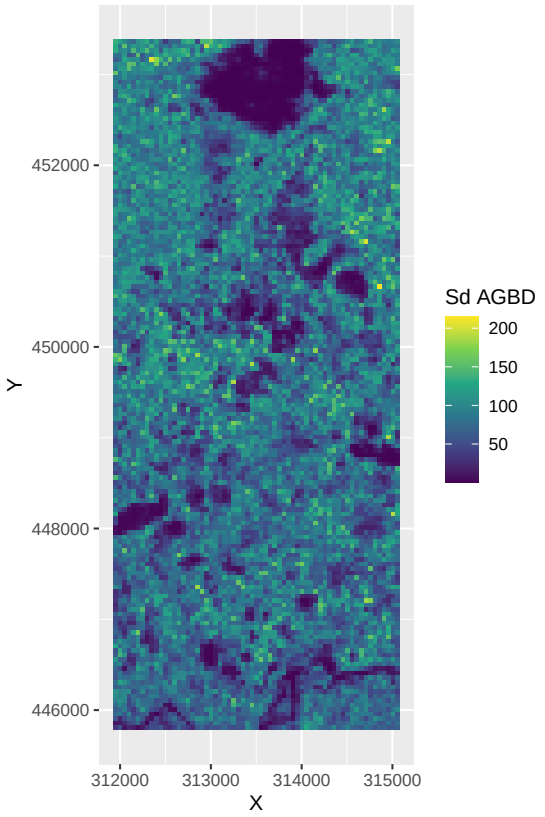
CHM-AGBD model calibration: example with Nouragues data



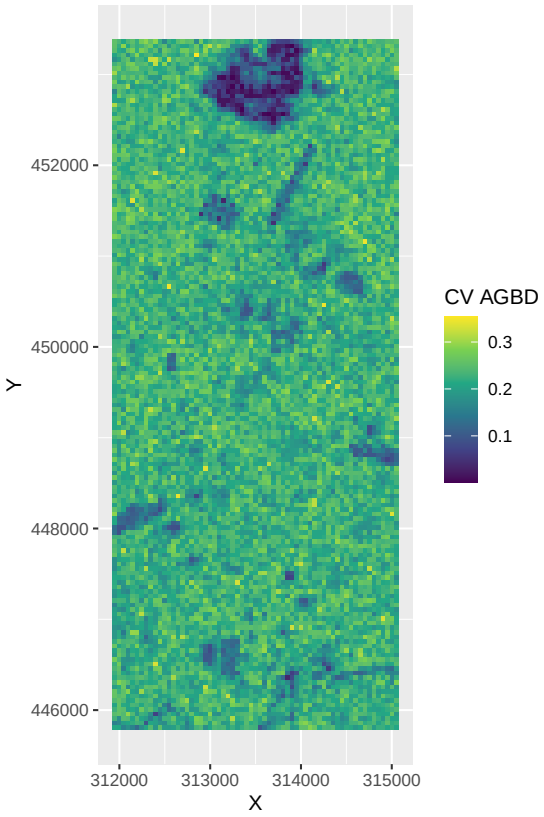
- SPV-C model
  $log(AGBD(s)) =$
  $(\beta + \tilde{\beta}(s)) \times log(CHM(s)) + \epsilon(s)$
  with
  $\tilde{\beta}(s_1), ..., \tilde{\beta}(s_n) \sim MVN(0, C_\beta(s_i, s_j))$

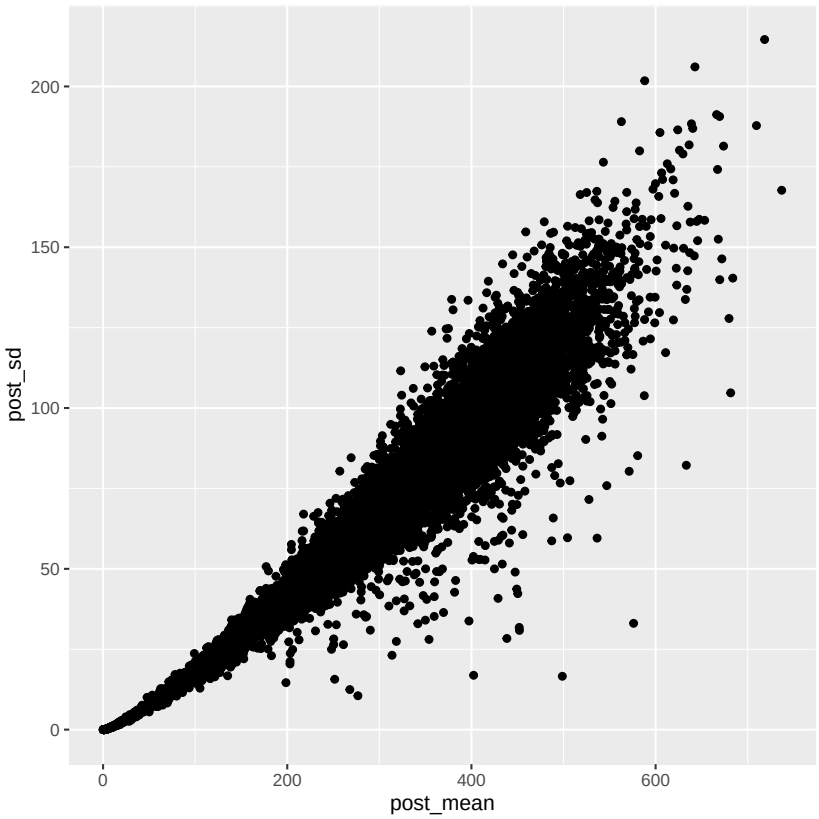
Predictions on Nouragues (brms)





Predictions on Nouragues: coefficient of variation (brms)





Two functions in the package

calibrate_model.R | predict_map.R

Source on Save | Run | Source

```r
#'  - 'x_center' and 'y_center': the coordinates of the plots/subplots in the projected coordinate system.
#'  - 'AGBD': the AGBD value of the simulation.
#'  - 'raster_metric': the raster metric value of the simulation.
#'
#' Speak about the model (fixed intercept), capturing the spatial autocorrelation, and cite Gelfand et al. 2003 (Spatial model
#'
#' If useCache = TRUE and this is the first time the model is being built, the model will be saved as a .rds file in the defin
#' If useCache = TRUE and the model has already been built using the user cache, the model will be loaded and updated to avoid
#' If useCache = NULL, the cache is first cleared before building the model.
#'
#' @export
#'
#' @author Arthur Bailly, Dominique Lamonica
#'
#' @importFrom data.table is.data.table copy setnames
#'
#' @examples
#' \dontrun{
#'
#' }

calibrate_model <- function(long_AGB_simu, nb_rep = 30, useCache = FALSE, plot_model = TRUE, chains = 4, thin = 20, iter = 230

  # Checking arguments -----------------------------------------------------


  # Check if package brms is available
  if (!requireNamespace("brms", quietly = TRUE)) {
    warning(
      'To build bayesian models, you must install the "brms" library \n\n',
      '\t\tinstall.packages("brms")'
    )
```

CHM-AGBD model calibration: implementation possibilities

- propagation of AGBD estimates uncertainties
- SPDE procedure
- future statistical development to use all the CHM spatial structure: better use of available information for a more robust & precise full spatial AGBD prediction (for a next major version)